

Titanic Survival Prediction Using Machine Learning

1. Introduction

The sinking of the RMS Titanic in 1912 is one of the most infamous maritime disasters in history. During this tragedy, more than 1500 passengers lost their lives. However, survival was not completely random; it depended on several factors such as gender, age, ticket class, and fare paid. This project aims to build a machine learning model that predicts whether a passenger survived the Titanic disaster based on available passenger information.

This project is a classic beginner-level machine learning problem and is widely used to understand data preprocessing, exploratory data analysis (EDA), feature engineering, model building, and evaluation.

2. Objective of the Project

The main objectives of this project are: - To understand and analyze the Titanic passenger dataset - To perform data cleaning and preprocessing - To build a machine learning model for survival prediction - To evaluate the performance of the model using standard metrics

3. Dataset Description

The Titanic dataset contains information about individual passengers. The dataset used in this project was loaded from the Seaborn library.

Key Features:

- **survived:** Survival status (0 = No, 1 = Yes)
 - **pclass:** Ticket class (1st, 2nd, 3rd)
 - **sex:** Gender of the passenger
 - **age:** Age of the passenger
 - **fare:** Fare paid for the ticket
 - **embarked:** Port of embarkation (C = Cherbourg, Q = Queenstown, S = Southampton)
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4. Tools and Technologies Used

- Programming Language: Python
- Environment: Jupyter Notebook
- Libraries:

- Pandas (data manipulation)
 - NumPy (numerical operations)
 - Matplotlib & Seaborn (data visualization)
 - Scikit-learn (machine learning)
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5. Data Preprocessing

5.1 Handling Missing Values

- The **age** column had missing values, which were filled using the median age.
- The **embarked** column had missing values, which were filled using the most frequent category (mode).

5.2 Encoding Categorical Variables

Machine learning models require numerical input. Therefore:

- The **sex** column was converted into numerical form using label encoding.
- The **embarked** column was also label encoded.

5.3 Feature Selection

Only relevant features were selected to reduce noise and improve model performance. The final feature set included:

- pclass, sex, age, fare, embarked

6. Exploratory Data Analysis (EDA)

Exploratory analysis helped in understanding relationships between features and survival:

- Female passengers had a significantly higher survival rate than males.
- Passengers in 1st class had better survival chances compared to 2nd and 3rd class.
- Younger passengers showed relatively higher survival probability.

EDA provided insights that justified the choice of features for model training.

7. Model Building

7.1 Model Selection

Logistic Regression was chosen because:

- It is suitable for binary classification problems
- It is easy to interpret
- It performs well on small and medium-sized datasets

7.2 Train-Test Split

The dataset was split into:

- 80% training data
- 20% testing data

This ensures unbiased evaluation of the model.

8. Model Evaluation

The trained model was evaluated using the following metrics:

8.1 Accuracy

Accuracy measures the percentage of correctly predicted instances.

8.2 Classification Report

The classification report includes: - Precision - Recall - F1-score

These metrics provide a detailed evaluation of model performance.

8.3 Confusion Matrix

A confusion matrix was used to visualize: - True Positives - True Negatives - False Positives - False Negatives

The results showed that the model performed reasonably well in predicting passenger survival.

9. Results and Discussion

The Logistic Regression model achieved satisfactory accuracy and demonstrated that survival on the Titanic was strongly influenced by gender, passenger class, and age. Although the model is simple, it effectively captures important patterns in the data.

10. Conclusion

In this project, a complete machine learning pipeline was implemented to predict Titanic passenger survival. The project covered data preprocessing, feature engineering, model training, and evaluation. This project helped in understanding the practical application of machine learning techniques on real-world data.

11. Future Scope

The project can be improved further by: - Using advanced models like Random Forest or Gradient Boosting - Performing hyperparameter tuning - Adding more engineered features - Using cross-validation for better evaluation

12. References

- Titanic Dataset (Seaborn Library)
- Scikit-learn Documentation
- Pandas Documentation

End of Report